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Gas Leakage Detection System



FINAL REPORT

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01) Overview

Liquefied Petroleum Gas (LPG) is a main source of fuel, especially in urban areas because it is clean compared to firewood and charcoal. Gas leakage is a major problem in the industrial sector, residential premises, etc. Nowadays, home security has become a major issue because of increasing gas leakage. Gas leakage is a source of great anxiety with ateliers, residential areas and vehicles like Compressed Natural Gas (CNG), buses, and cars which are run on gas power. One of the preventive methods to stop accidents associated with the gas leakage is to install a gas leakage detection kit at vulnerable places. The aim of this project is to propose and discuss a design of a gas leakage detection system that can automatically detect, alert and control gas leakage. This

Proposed system also includes an alerting system for the users. The system is based on a sensor that easily detects a gas leakage.

02) Introduction

Gas leakage is a serious problem and nowadays it is observed in many places like residences, industries, and vehicles like Compressed Natural Gas (CNG), buses, cars, etc. It is noticed that due to gas leakage, dangerous accidents occur. Liquid petroleum gas (LPG) is highly inflammable and can burn even at some distance from the source of leakage. This energy source is primarily composed of propane and butane which are highly flammable chemical compounds. These gases can catch fire easily.

In homes, LPG is used mainly for cooking purposes. When a leak occurs, the leaked gases may lead to an explosion. Gas leakage leads to various accidents resulting in both material loss and human injuries. Home fires have been occurring frequently and the threat to human lives and properties has been growing in recent years. The risks of explosion, fire, suffocation are based on their physical properties such toxicity, flammability, etc. The number of deaths due to the explosion of gas cylinders has been increasing in recent years. Approximately 847 gas accidents have been reported in 2021. The reason for such explosions is due to substandard cylinders, old valves, no regular checking of gas cylinders, worn out regulators and a lack of awareness of handling gas cylinders. Therefore, the gas leakage should be detected and controlled to protect people from danger.

A gas leakage detector becomes vital and helps to protect people from the dangers of gas leakage. In this project a low-cost advanced sensor-based gas leakage detector, alert and control system is proposed and discussed. The system is very efficient, user friendly, portable, small in size and cost effective.

03) Aim

The main aim of this project is to avoid the accident and death in the gas leakage explosion in house, hotels and industries. Domestically we use natural gas and it is very useful for burning purpose. If this gas is leaked in our kitchens, hotels or factories and not sensed in time, it may lead to fatal disaster, and may cause human and material loss. For this purpose we have developed “GAS LEAKAGE DETECTION SYSTEM”.

04) Objective

The objective of this project is to detect any leakage of LPG/CNG based cars, small scale factories or in home appliances also. It will detect the leakage and will close the Knob of the system to stop the supply of the gas. Stepper motor is there that could be attached with the knob and will close the supply by rotating it. An alarm is there also the stop the alert the user as soon as leakage is found.

05) Methodology

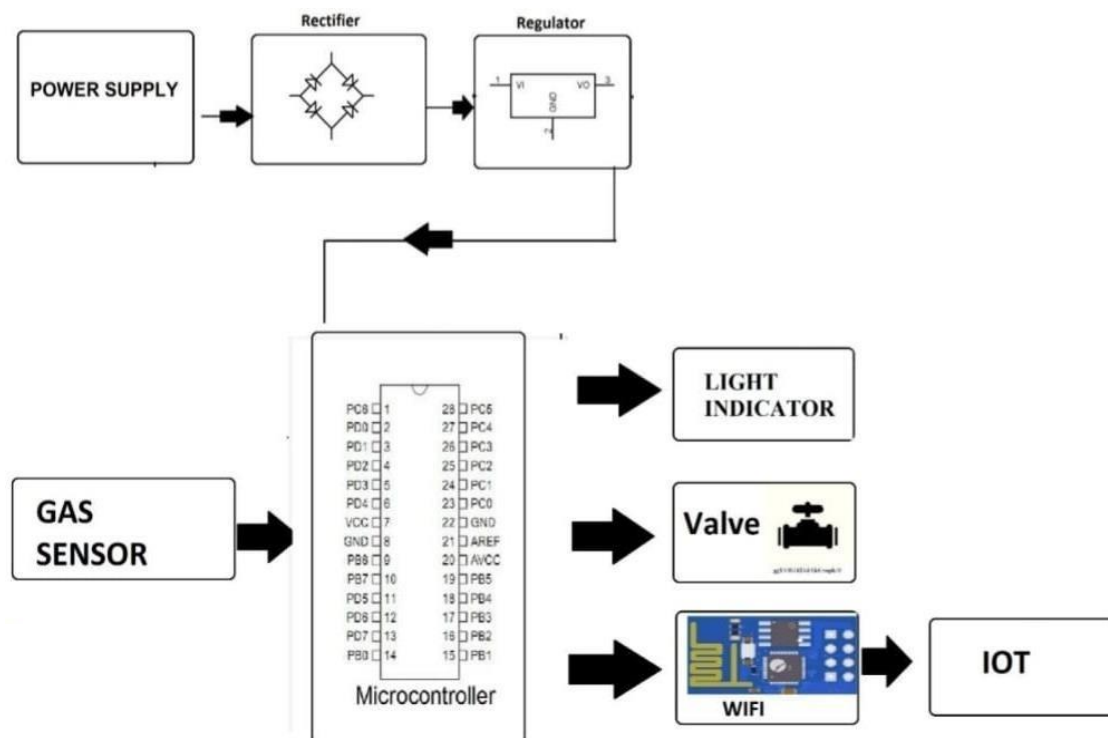
First we have to find a sensor to check whether the LP gas is being in the atmosphere. The signal that we have receive from the sensor is given to the arduino circuit board as an input. By using that input we suggest to make an output to the speaker and to relay module by coding. Then that will add to the chip in circuit board and made the system. After that a servo motor is connected across the relay module and by using that servo motor we wish to make an accessory to energizing the faucet which is placed between to gas stove and gas pipe. Finally power current will supplied to the circuit board by using power supply and complete the circuit.

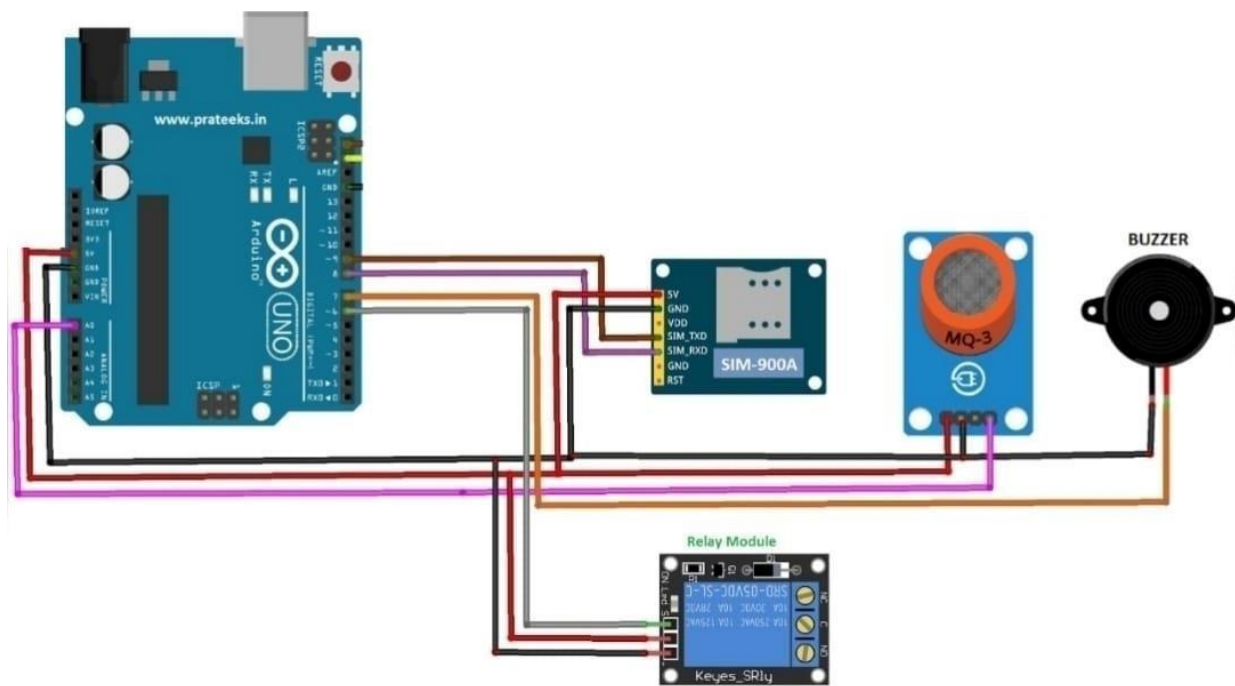
Now if the gas get a leakage the sensor will identify it and then microcontroller in circuit will work as according to the source code that we already given and then the gas supply will off and the speaker will out the sound to get that the gas leakage to the user and using SIM-900A can get a message to our mobile phone.

06)Time Plan

Task	Aug 2022	Sep 2022	Oct 2022	Nov 2022	Dec 2022
Topic selection and submission of proposal					
Literature survey					
Gathering required information for the proposed system					
Design and develop the system					
System simulation					
Reflective Log 1					
Progress report 1					
Progress presentation 1					
Final report submission					

09)Structure





10)Budget Plan

Arduino board Uno (DB002)	Rs 4 300
Servo motor	Rs 1 400
Signal warning light flash siren LED lamp	Rs 2 600
Alarm - buzzer module	Rs 180
Wi-Fi module - SIM900A	Rs 2 250
UPS = 650VA	Rs 5 000
Power Supply (12/1A)	Rs 1 250
Gas Sensors MQ-5	Rs 750
Total	Rs 17 730

Conclusion

The design of a sensor-based automatic gas leakage detector with an alert and control system has been proposed and discussed in this project. This is a low-cost, low power, lightweight, portable, safe, user friendly, efficient, multi featured and simple system device for detecting gas. Gas leakage detection will not only provide us with significance in the health department but it will also lead to raise our economy, because when gas leaks it not only contaminates the atmosphere but also wastage of gases will hurt our economy. In the open literatures it is noticed that much work has not been

Done for a smart gas detection system. In future, more advanced features will be integrated with this system which will provide users with more safety and relaxation. The proliferation of handheld devices has led to developments in the field of smart gas sensors, which has considerably widened their scope of application. The need for ensuring safety in workplaces is expected to be the key driving force for the market over the coming years.
